◆ TECHNICAL BULLETIN

Hull Material Integrity Ratings and Ballistic Breach Risk Classification

Docking Ring District, Commercial Zones, and Exterior Corridor Infrastructure
Vesta Station — Cooperative Engineering Authority

VSE-TB-0291

ISSUED	AUGUST 2113
AF DATE	9 AF
REVISION	1.0
STATUS	ACTIVE
SUPERSEDES	VSE-TB-0244
NEXT REVIEW	AUGUST 2115

◆ 1. PURPOSE AND SCOPE

This bulletin establishes **standardised damage resistance (DR) ratings** for hull and structural materials across Vesta Station's primary occupation zones. It is issued in response to a pattern of incidents in the 2111–2112 operational period involving high-velocity projectile impact on exterior-facing composite panels in the Docking Ring District. Ratings given herein supersede those in VSE-TB-0244 (issued 2109) and reflect updated material specifications following the Station's partial infrastructure renewal programme.

This bulletin applies to all station personnel, visiting crew, commercial operators, and licensed security contractors operating within Vesta Station's jurisdiction. Compliance with REG04-23-FIREARMS is mandatory in all zones where composite or transparisteel materials form part of the pressure boundary.

◆ 2. INCIDENT BASIS

The following incident records prompted revision of the 2109 ratings and the introduction of mandatory breach-risk signage in composite zones:

INCIDENT VSE-INC-2112-047 // 14 FEBRUARY 2112 // DOCKING RING, EXTERIOR CORRIDOR 4-F

Gravel-class micrometeorite (est. 4–6mm diameter, impact velocity approx. 12 km/s) penetrated transparisteel viewport panel TP-4F-09 at the lower sill seam. Full perforation confirmed. Pressure loss: 0.003 atm/hr. Detected by atmospheric monitoring at T+22 minutes. Sealed by maintenance droid at T+41 minutes. No casualties. Panel rated DR 12 at time of incident. **Rating revised to DR 10 following metallurgical analysis of impact site.**

INCIDENT VSE-INC-2112-091 // 3 JUNE 2112 // DOCKING RING, MODULE JUNCTION 4-D/4-E

Discharge of semi-automatic kinetic pistol (single round, medium calibre) during a commercial dispute. Round impacted exterior-facing plastisteel partition panel PP-4DE-03. Partial penetration to 60% panel depth. No breach. Hairline fracture propagated 14cm from impact point. Panel replaced. Incident prompted formal review of ballistic resistance ratings for composite panels in commercial zones. **Discharge in this zone constituted a violation of REG04-23-FIREARMS.** Operator charged.

INCIDENT VSE-INC-2112-114 // 29 AUGUST 2112 // DOCKING RING, VIEWPORT ARRAY 4-G

Maintenance droid (unit M-DR-0047) recorded seventeen (17) existing micrometeorite impact sites during scheduled inspection of viewport array 4-G, of which nine (9) showed full perforation with historic sealant caulking. Six (6) sites had not been previously logged. Sealant integrity on three (3) sites rated marginal. All sites resealed. **Inspection frequency for exterior viewport arrays increased from 180-day to 60-day cycle.**

◆ 3. ZONE CLASSIFICATION AND MATERIAL SUMMARY

Vesta Station infrastructure falls into three primary structural categories for the purpose of ballistic and impact risk assessment:

ZONE A — REGOLITH (CARVED)

Raw consolidated basaltic asteroid rock. Applies to all tunnel corridors, residential hab blocks, carved service passages, and sub-surface commercial spaces. Thickness: typically 8–40 metres between occupied space and exterior surface.

DR: EFFECTIVE INFINITE

ZONE B — PLASTISTEEL COMPOSITE

Structural composite panels used in prefabricated modules, docking ring outer-face sections, commercial zone partitions, and interior bulkheads with pressure-boundary function. Rated for sustained vacuum exposure.

DR: 20–30 (panel-dependent)

ZONE C — TRANSPARISTEEL COMPOSITE

Optically clear structural composite used in viewports, observation panels, and corridor windows with exterior-facing function. Thinner than structural plastisteel by design necessity. Subject to micrometeorite impact.

DR: 10–15 (viewport-dependent)

ZONE D — DOCKING RING OUTER HULL

Layered plastisteel with vacuum insulation. Forms exterior hull face of docking ring. Thicker than standard partition panels. Breach risk is primarily gasket and seal failure rather than panel perforation. Impact may compromise seal integrity without full panel perforation.

DR: 25–35 (section-dependent)

◆ 4. MATERIAL DR RATINGS — BALLISTIC IMPACT REFERENCE

MATERIAL / LOCATION	ZONE	DR RATING	BREACH RISK	NOTES
Basaltic regolith (carved tunnel)	A	∞	NIL	No ballistic risk to structural integrity. Ricochet and spall hazard to occupants.
Plastisteel partition (interior, non-pressure)	B	30	LOW	No vacuum boundary function. Structural damage only.
Plastisteel panel (exterior-facing, pressure boundary)	B	20–28	MODERATE	Single AP round at close range may perforate. Sustained fire will breach. See VSE-INC-2112-091.
Transparisteel viewport (standard)	C	12	HIGH	Micrometeorite incidents confirmed at this rating. Medium pistol round will breach. Flechette pattern automatic breach. See VSE-INC-2112-047.
Transparisteel viewport (reinforced, airlock zones)	C	20	MODERATE	Airlock and docking interface viewports only. Heavier gauge. Not present in general corridor sections.
Viewport sill seam / gasket	C	6–8	CRITICAL	Seam is the weakest point of any viewport assembly. Impact at sill seam may cause breach even if panel face is not perforated. Flechette weapons targeting any viewport should be treated as seam-impact events.
Docking ring outer hull (panel face)	D	30	MODERATE	Panel face resists most small arms. Sustained fire will breach. Gasket failure risk precedes panel perforation.
Docking ring outer hull (panel gasket / seal)	D	10	HIGH	Gasket is rated separately from panel. Pinhole breach via gasket failure will not trigger hull alarm but will show on atmospheric monitoring within 15–30 minutes.

! ◈ SPECIAL NOTICE — FLECHETTE AND BURST-GROUP WEAPONS

Flechette-pattern weapons (including shredder-class and multi-flechette discharge systems) present a **categorically elevated breach risk** in all Zone B, C, and D environments. The spread pattern of a standard flechette discharge at combat range (2–8 metres) exceeds the face area of a standard viewport panel and will impact the sill seam regardless of aim point. **A flechette discharge anywhere in the vicinity of a viewport assembly should be treated as a guaranteed seam-impact event.**

This is the primary basis for the single-shot lock requirement under **REG04-23-FIREARMS**. The regulation does not assume malicious intent — it assumes that in a high-stress situation, a flechette weapon on burst setting fired in a Zone B/C/D environment will cause a breach regardless of the operator's intentions.

Station security is authorised to treat an unlocked flechette weapon in a composite zone as an immediate hull-breach threat and respond accordingly.

◈ 5. BREACH EVENT RESPONSE

In the event of confirmed or suspected hull breach in any zone:

Slow leak (DR 10–15 perforation, gasket failure): Atmospheric monitoring will detect pressure drop within 15–30 minutes. Pressure drop alarm triggers automatic notification to station security and Engineering Authority dispatch. Maintenance droid response time: 8–15 minutes from alarm. Occupants should evacuate the affected corridor section and seal bulkhead doors at either end.

Fast leak (multiple perforations, large panel failure): Audible pressure differential within seconds. Emergency bulkhead lockdown triggers automatically at 0.02 atm/min drop rate. Do not attempt to manually seal. Evacuate immediately toward the nearest regolith zone — Zone A sections will not be affected by pressure loss.

Discharge incidents causing breach: Any person discharging a weapon that causes or contributes to a hull breach is liable under Vesta Station Cooperative Law, Section 44 (Reckless Endangerment of Station Integrity). Engineering Authority incident response will preserve discharge evidence for legal proceedings.

VSE

CHIEF STRUCTURAL ENGINEER
VESTA STATION ENG. AUTHORITY

COOPERATIVE COUNCIL LIAISON
COMPLIANCE SIGN-OFF

SECURITY AUTHORITY REVIEW
REG04-23-FIREARMS COMPLIANCE

PUBLIC POSTING MANDATORY